Investigating the effect of pH level of simulated pool water on chlorine oxidation kinetics

1. Introduction

There is a swimming pool in the backyard of my house. I spent a lot of time maintaining its water chemistry. The most important part of this process is adding sanitizer to control bacteria, biotics, algae and other contaminants to keep the pool clean and safe for swimmers. The most common sanitizer in swimming pools is chlorine sanitizer, namely sodium or calcium hypochlorite. It is also common in household cleaners because of its strong oxidizing power and availability. However, to make chlorine work efficiently in swimming pools, pool maintenance instructions always remind pool owners to “adjust pH before adding chlorine”(Editorial Team, 2017). Because of this, every time I tested the pH level before adding chlorine, I felt confused about the reasoning behind this. I did some research and designed an experiment to understand this phenomenon.

2. Background Information

2.1 The Working Mechanism of Chlorine Sanitizers

Chlorine sanitizers, usually referred to as sodium hypochlorite or calcium hypochlorite, are commonly used as household cleaners or swimming pool sanitizers. Sodium hypochlorite is usually called bleach and presented in a liquid state. It is effective in cleaning stains and disinfecting. Calcium hypochlorite is mostly used as chlorine sanitizers in swimming pools and usually presented as powders or tablets. Both chlorine sanitizers produce the same substance in water solution: hypochlorous acid, a strong oxidizing agent.

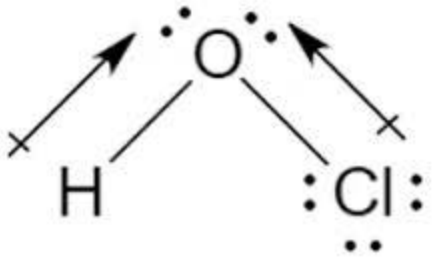


Figure 1: Structural formula of hypochlorous acid(Honeydew Labs. n.d.)

Because the strong electronegativity of oxygen, it pulls the shared electron pair with the chlorine atom towards itself, causing the chlorine atom to have a partial positive charge(δ^+). This makes the chlorine atom behave as an electrophilic center that attracts electrons from lone pairs(Honeydew Labs, n.d.) and π bonds. It also oxidizes amino groups, disrupts enzyme formation, and destroys cell membranes of microbes(Pailakian A, 2024 May 14).

2.2 Effect of pH on Hypochlorous Acid and Hypochlorite Ions

Here is how it is produced from chlorine sanitizers:

When sodium or calcium hypochlorite is dissolved in water:

$$NaOCl(aq) \rightarrow Na^{+}(aq) + OCl^{-}(aq)$$
$$Ca(OCl)_2 \rightarrow Ca^{2+}(aq) + 2OCl^{-}(aq)$$

Then, hypochlorite ions undergo base hydrolysis reaction in water:

$$OCl^{-}(aq) + H_2O(l) \Leftrightarrow HOCl(aq) + OH^{-}(aq)$$

The K_b value of this reaction is 3.3×10^{-7}(Brown T, 2020), meaning the reaction is reactant-favored.

However, some hypochlorous acid is still produced. Then it undergoes acid dissociation reaction in water:

$$HOCl(aq) + H_2O(l) \Leftrightarrow H_3O^+(aq) + OCl^-(aq)$$

The K_a value of this reaction is 3.0×10^{-8}(Brown T, 2020), meaning the reaction is reactant-favored. However, when the pH level increases, meaning the concentration of hydroxide ions increases, the hydronium ions are neutralized therefore its concentration decreases. According to Le Chatelier's Principle, the reaction shifts to the product side, which increases the concentration of hypochlorite ions. When pH level decreases, the concentration of hydronium ions increases, the reaction shifts to the reactant side, which increases the concentration of hypochlorous acid.

The relative amount of hypochlorous acid and hypochlorite ion largely determines the effectiveness of the sanitizer. Both of them contain a bond between an electronegative oxygen and a partially positively charged chlorine, however hypochlorite ion is a relatively weaker oxidizing agent than hypochlorous acid because it repels electrons due to its negative charge. Therefore, pH level is crucial for chlorine sanitizer to work properly in swimming pools.

2.3 Lab Design

To investigate this topic while still partially representing a real world scenario, for each research sample, simulate a pool water environment by adjusting other important parameters: total alkalinity, calcium hardness and salt. Total alkalinity shows the amount of buffer present in the water and is usually raised by adding sodium bicarbonate. It resists the pH adjustment of the experiment. Calcium hardness shows the amount of calcium present in the water, its primary purpose is to protect swimming pool liners and equipment. Calcium hardness level is raised by

adding calcium chloride. These substances act as parameters that only create a realistic simulation, while not affecting the main purpose of this experiment.

I adjusted pH level from 6 to 9 with 0.5 difference for each trial. In real swimming pool maintenance, the acceptable and realistic pH range of pool water is usually 7.2~7.8. However, a larger pH range was chosen for observing clearer trends of the effects, which allows comprehensive examination of the relationship between pH level and sanitizer efficiency.

Measuring the rate of the reaction between chlorine sanitizer and the target substance is a practical way to determine “chlorine effectiveness” quantitatively. A dye can be used as the target substance based on its chemical properties. Many dyes contain a conjugated π system with high electron density for oxidation reactions(Wood A, n.d.). When those π systems are disrupted by oxidation, the dye’s ability to absorb light is reduced, causing a change in color. The absorbance of color of the dye can be quantitatively measured and recorded by a colorimeter.

According to Beer-Lambert Law(Libretexts, 2023):

$$A = \varepsilon l c$$

A: Absorbance

ε: Molar absorptivity($L \cdot mol^{-1} \cdot cm^{-1}$)

l: Length of light path(cm)

c: Concentration($mol \cdot L^{-1}$)

Since the path and wavelength are kept constant for all measurements, absorbance is directly proportional to dye concentration. Therefore, the absorbance over time data reflects the decrease in the concentration due to oxidation. Thus, the slope is proportional to the reaction rate.

Based on my topic setups, I chose indigo carmine as the target substance for oxidation. It is a common redox indicator(Agrawal S, 2025), which changes color easily during oxidation reactions. The color of indigo carmine solution is blue, which means it primarily absorbs light within the red-orange wavelength region. Therefore when measuring the red light absorbance of each sample, the reaction rate can be clearly observed by the colorimeter.

3. Hypothesis

When the pH level is decreased, absorbance of indigo carmine vs time graph measured by colorimeter is expected to decline rapidly, resulting in generally steeper slopes along the curve. When the pH level is increased, absorbance of indigo carmine vs time graph measured by colorimeter is expected to decline more slowly, resulting in generally shallower slopes along the curve.

4. Variables

4.1 Independent and Dependent Variables

Independent variable:

pH level of each sample of simulated pool water(6.0, 6.5, 7.0, 7.5, 8.0, 8.5, 9.0). It is measured by a PASCO Wireless pH Sensor. Adjusted by 0.1 M hydrochloric acid(pH decreaser) and 0.1 M sodium carbonate solution(pH increaser).

Dependent variable:

The reaction rate of the oxidation of indigo carmine, measured by a PASCO Wireless Colorimeter.

4.2 Controlled Variables

Controlled Variables	Control Method(s)	Significance
The initial chemical composition of the simulated pool water	Prepared 1 L simulated pool water stock and all subsequent samples were taken from this stock.	Avoid factors other than pH level affecting the experiment purpose.
Volume of simulated pool water taken	For every pH level chosen, 100 mL stock is taken for subsequent procedures.	Ensures the concentration of chlorine sanitizer added to the sample is the same.
Concentration of chlorine sanitizer added	Prepared 1 L diluted bleach stock and is the only source of chlorine sanitizer in all	Ensures the same chlorine concentration for oxidizing indigo carmine.

	subsequent procedures.	
Amount of chlorine sanitizer added	For every pH level chosen, 0.5 mL stock of diluted bleach is added to every 100 mL simulated pool water.	Ensures the same chlorine concentration for oxidizing indigo carmine.
Concentration of indigo carmine solution added	Prepared 1 L diluted indigo carmine solution and is the only source of indigo carmine in all subsequent procedures.	Ensures the same amount of indigo carmine for each sample.
Amount of indigo carmine solution added	For every pH level chosen, 4 mL of diluted indigo carmine solution is added to every 50 mL chlorinated sample.	Ensures the same amount of indigo carmine for each sample.
External environment	The whole experiment was taken under standard and stable laboratory conditions.	Avoid factors that may affect experiment outcomes such as temperature.
Storage of chemicals	All chemicals that weren't used immediately after being made were sealed in	Avoid reactant loss due to external factors such as evaporation or degradation

	volumetric flasks or reagent bottles. Specifically for bleach stock, the volumetric flask that stored it was fully covered by opaque brown paper.	by sun.
Order of mixing chemicals	The same steps were being used for preparing the simulated pool water, diluted bleach, and diluted indigo carmine solution. (Section 4.2)	Avoid different orders of adding chemicals affecting experiment outcomes.
Measurement method	The same pH sensor and colorimeter were used throughout the experiment and were properly setted and calibrated.	Minimize the uncertainty in measurement accuracy.

4.3 Uncontrolled Variables

The first uncontrolled variable was the extent of diffusion of diluted indigo carmine solution into chlorinated simulated pool water. To allow the colorimeter to detect trends in the oxidizing

process of indigo carmine as early as possible, the mixing process is controlled to be about 3~5 seconds. This was caused by insufficient mixing between the two solutions. It affected the starting concentration of indigo carmine in each sample, making different initial absorbance values.

The second uncontrolled variable was the reaction time before each measuring process started. Because the colorimeter only requires 3~4 mL of each sample, reactions already started before moving that small portion of sample into the cuvette. The amount of time taken before applying the colorimeter was controlled to be within 4 seconds and similar, but the exact time was uncertain. This variable caused a removal of some early data.

The third uncontrolled variable was the amount of initial chemical composition of the simulated pool water after pH adjustment. This was caused by the side effects of adding diluted hydrochloric acid and alkaline solution. For example, adjusting the pH level partially consumes the pH buffer in each sample. But its effect on the outcomes is negligible because the initial chemicals added to simulate pool water didn't affect the main reaction.

5. Experiment Design

5.1 Apparatus

1. Three 1 L volumetric flasks
2. Seven 250 mL volumetric flasks
3. One 100 mL volumetric flask

4. One 1 L reagent bottle
5. Two 50 mL burette
6. Two burette stand
7. Two 250 mL beakers
8. Two 50 mL beakers
9. Stirring rod
10. Funnel
11. Two droppers
12. 10 mL graduated cylinder
13. 50 mL graduated cylinder
14. 100 mL graduated cylinder
15. Calibration buffer solutions for pH sensor(pH 4.0 and 10.0)
16. Weighing boat
17. Microspatula
18. Analytical balance
19. PASCO Wireless pH Sensor
20. PASCO Wireless Colorimeter
21. Baking soda(Sodium Bicarbonate), 0.22 g per 1 L
22. Calcium chloride dihydrate, 0.29 g per 1 L
23. Sodium chloride, 3 g per 1 L
24. Hydrochloric acid, 8.3 mL per 1 L
25. Sodium carbonate, 1.06 g per 100 mL

26. Concentrated bleach, initially 19 mL Great Value Concentrated Bleach used.
27. Indigo carmine powder, initially 2 g used.
28. Distilled water

5.2 Method

- Add an appropriate amount of distilled water into a 250 mL beaker and dissolve baking soda, calcium chloride dihydrate, and sodium chloride in it.
- After stirring, pour the solution into a 1 L volumetric flask and dilute the solution into 1 L by adding distilled water until reaching the calibration line. This solution is the simulated pool water.
- Dilute concentrated(12 M) hydrochloric acid into 0.1 M by using a 1 L volumetric flask. This solution is the pH decreaser for adjusting the pH level of simulated pool water.
- Add an appropriate amount of distilled water into a 250 mL beaker and dissolve sodium carbonate in it.
- After stirring, pour the solution into a 100 mL volumetric flask and dilute the solution into 100 mL by adding distilled water until reaching the calibration line. This solution is the pH increaser for adjusting pH level of simulated pool water.
- Preparing seven volumetric flasks for storing simulated pool waters with different pH levels. Use two 50 mL beakers to store small amounts of pH increaser and decreaser for pH adjustment.
    - Use a 100 mL graduated cylinder to take out 100 mL simulated pool water.
    - Pour it into a 250 mL beaker.
    - Put the calibrated pH sensor into the solution to measure real-time pH value.

    - Start adjusting the pH level by adding the pH increaser and decreaser into the solution using two droppers and constantly stirring the solution.
    - When reaching the target pH level, the solution is stored in a volumetric flask.
    - Repeat the process for the next target pH level.
- Measure 19 mL concentrated bleach in a 10 mL graduated cylinder and dilute it into 1 L solution by adding distilled water. This solution is the chlorine sanitizer.
- Measure 2 g indigo carmine powder in a weighing boat by an analytical balance and dilute it by 100 mL distilled water in a 100 mL graduated cylinder.
- Take 10 mL of this solution by a 10 mL graduated cylinder and dilute it into 1 L solution by adding distilled water in a volumetric flask. This solution is the diluted indigo carmine solution.
- Set two 50 mL burettes and add 50 mL of chlorine sanitizer and diluted indigo carmine solution to each of them through a funnel.
- Start data collection by maintaining 5 ppm of chlorine sanitizer in the simulated pool water and test the rate of the reaction with diluted indigo carmine solution.
    - Transfer the simulated pool water with a specific pH level into a 250 mL beaker and add chlorine sanitizer into it through the burette.
    - After stirring, measure 50 mL of this sample by a 50 mL graduated cylinder.
    - Add the diluted indigo carmine solution to the graduated cylinder.
    - After mixing for a few seconds, take 3~4 mL of the sample into a cuvette and put into the colorimeter.

    - The colorimeter will output an Absorbance vs Time graph, which represents the rate of reaction of the oxidation of indigo carmine. This step is three minutes long.
    - Repeat this process for each specific pH level.

5.3 Safety and Ethical Assessment

Safety:

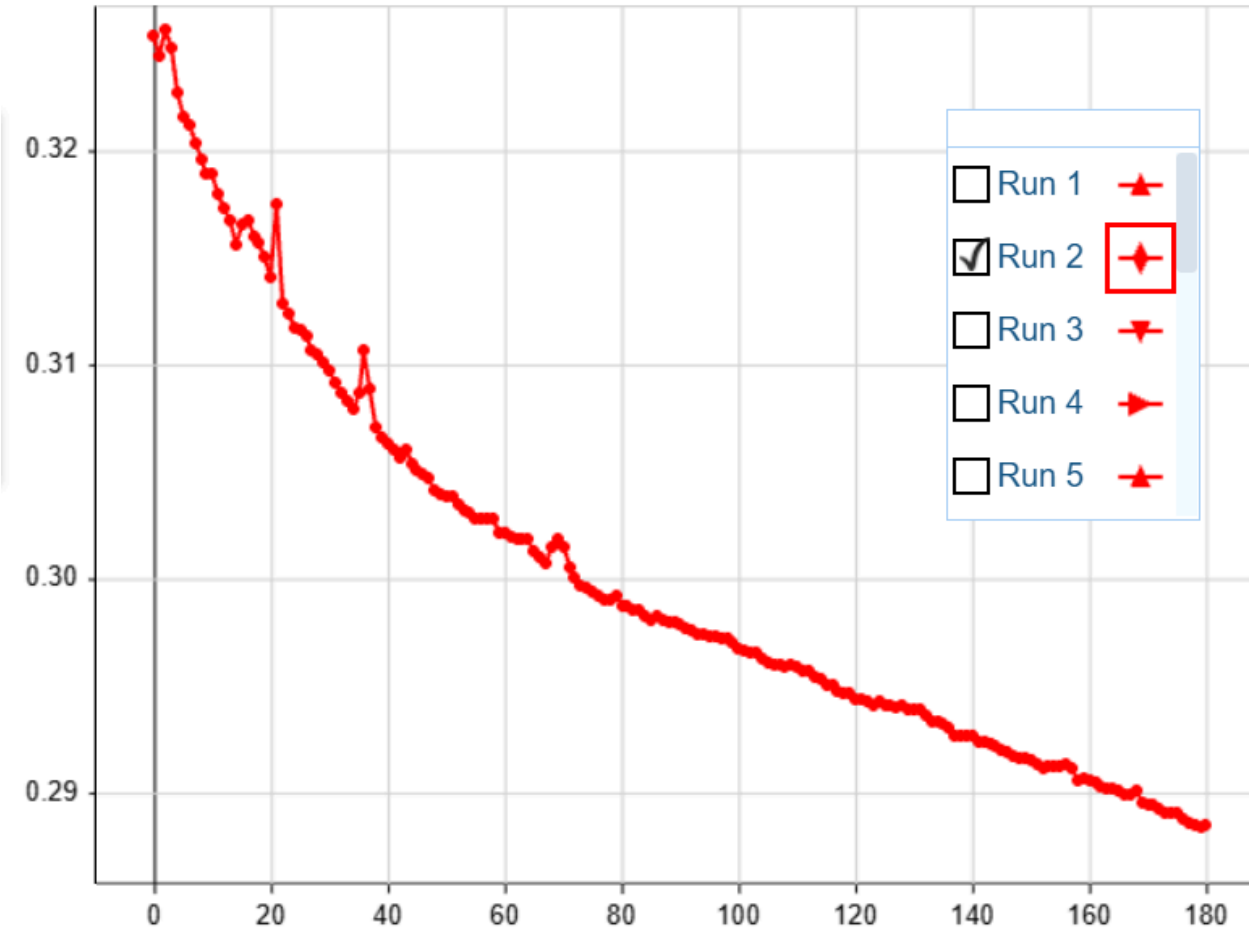
- The experiment involves making pH decreaser using 12 M hydrochloric acid. This step was completed under the supervision of a teacher, and protective equipment was worn throughout.
- This experiment involves multiple chemicals with potential danger, including acidic and basic solutions, concentrated and diluted bleach. Safety goggles were worn throughout.

Ethical and environmental:

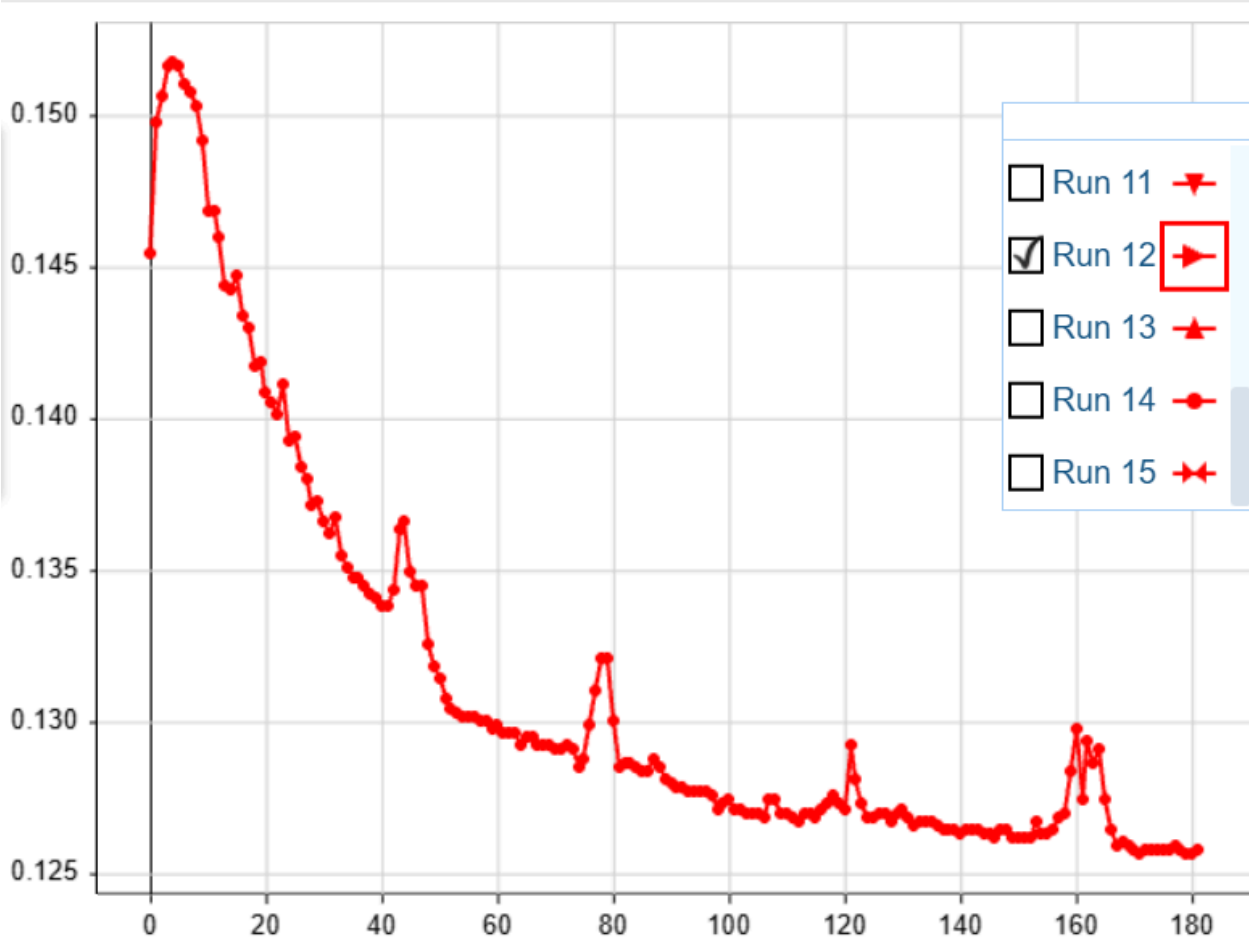
The chemicals produced and extra materials in this experiment were disposed of properly.

6. Data analysis

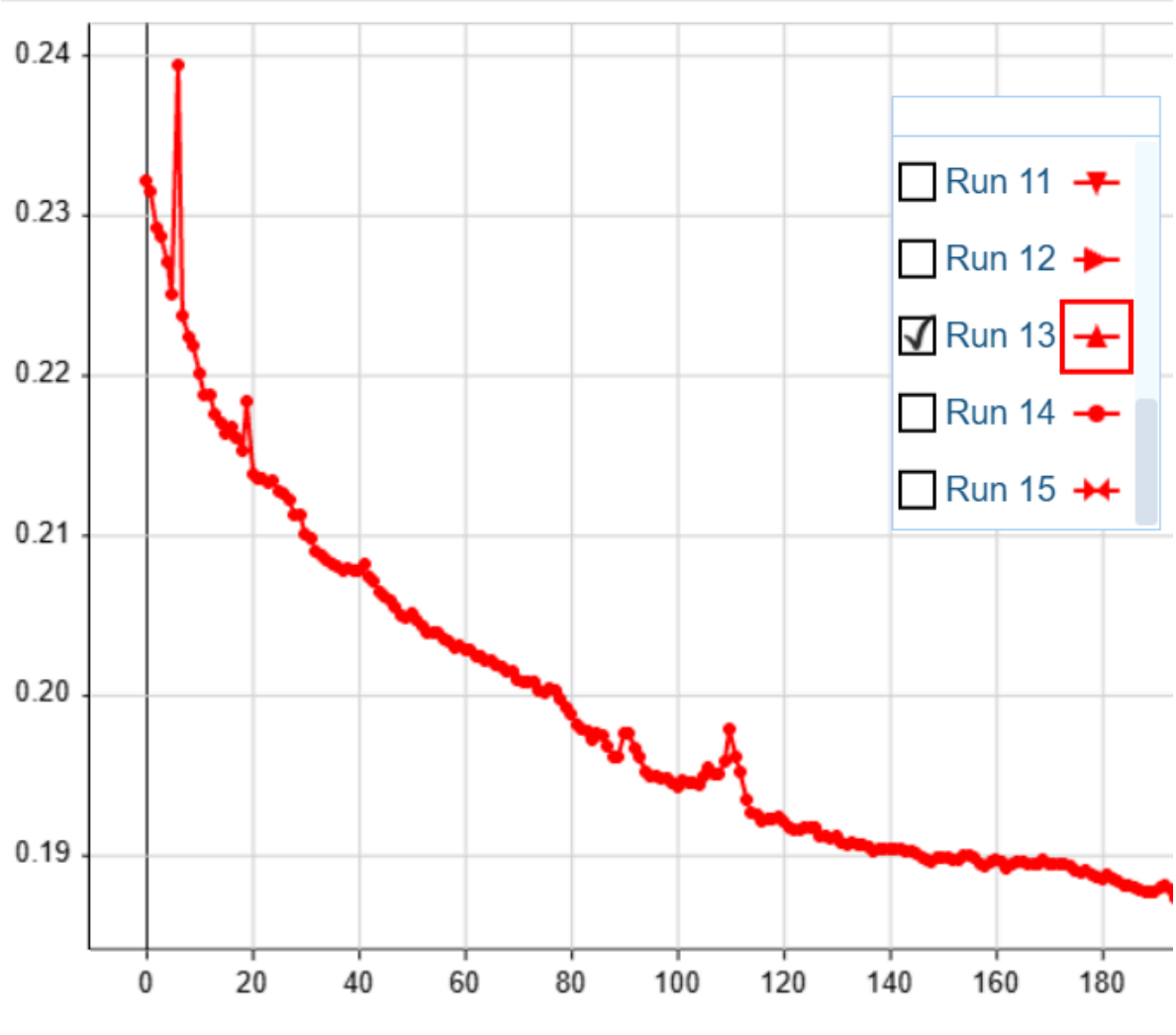
6.1 Raw Data



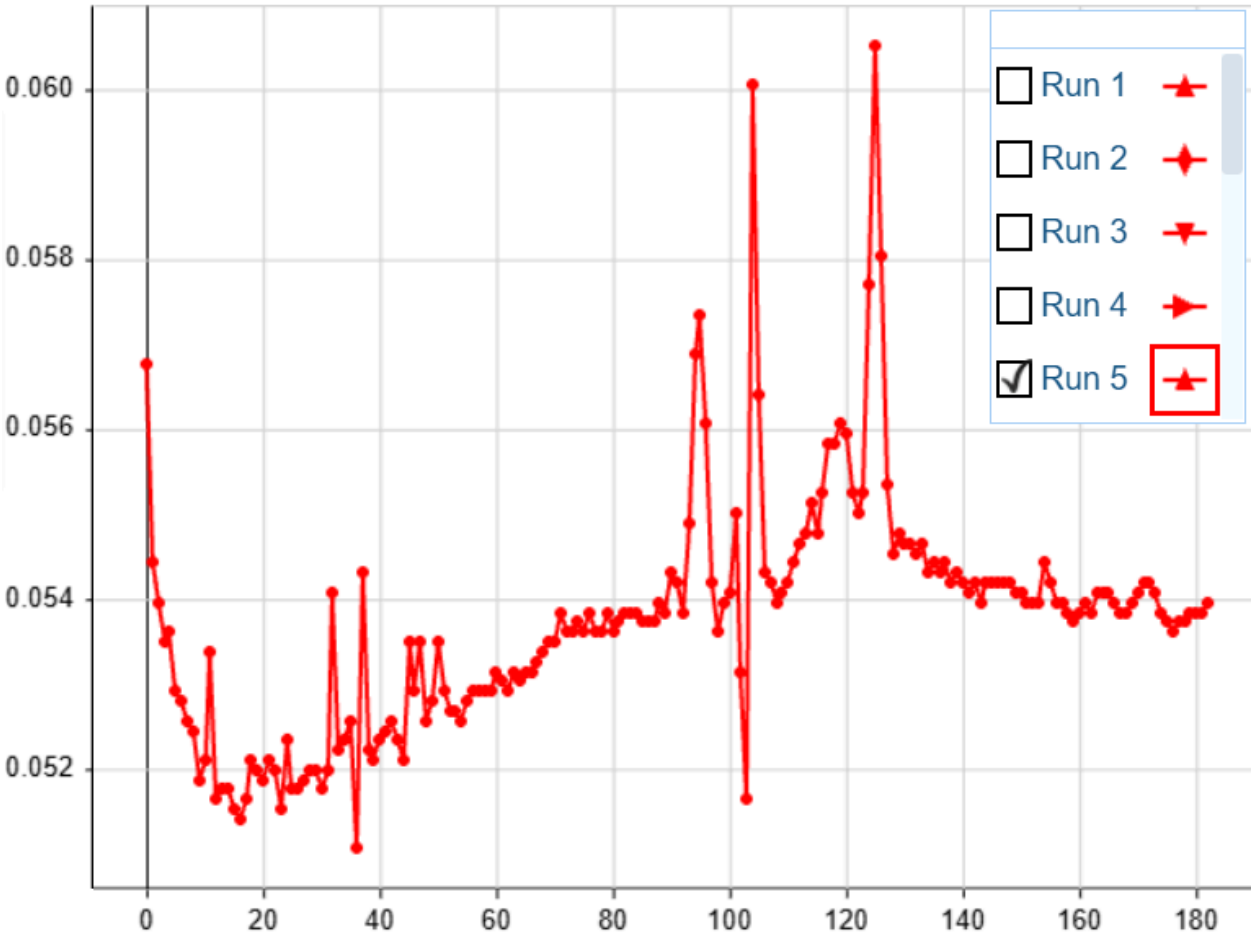
Graph 1: at pH 7.5



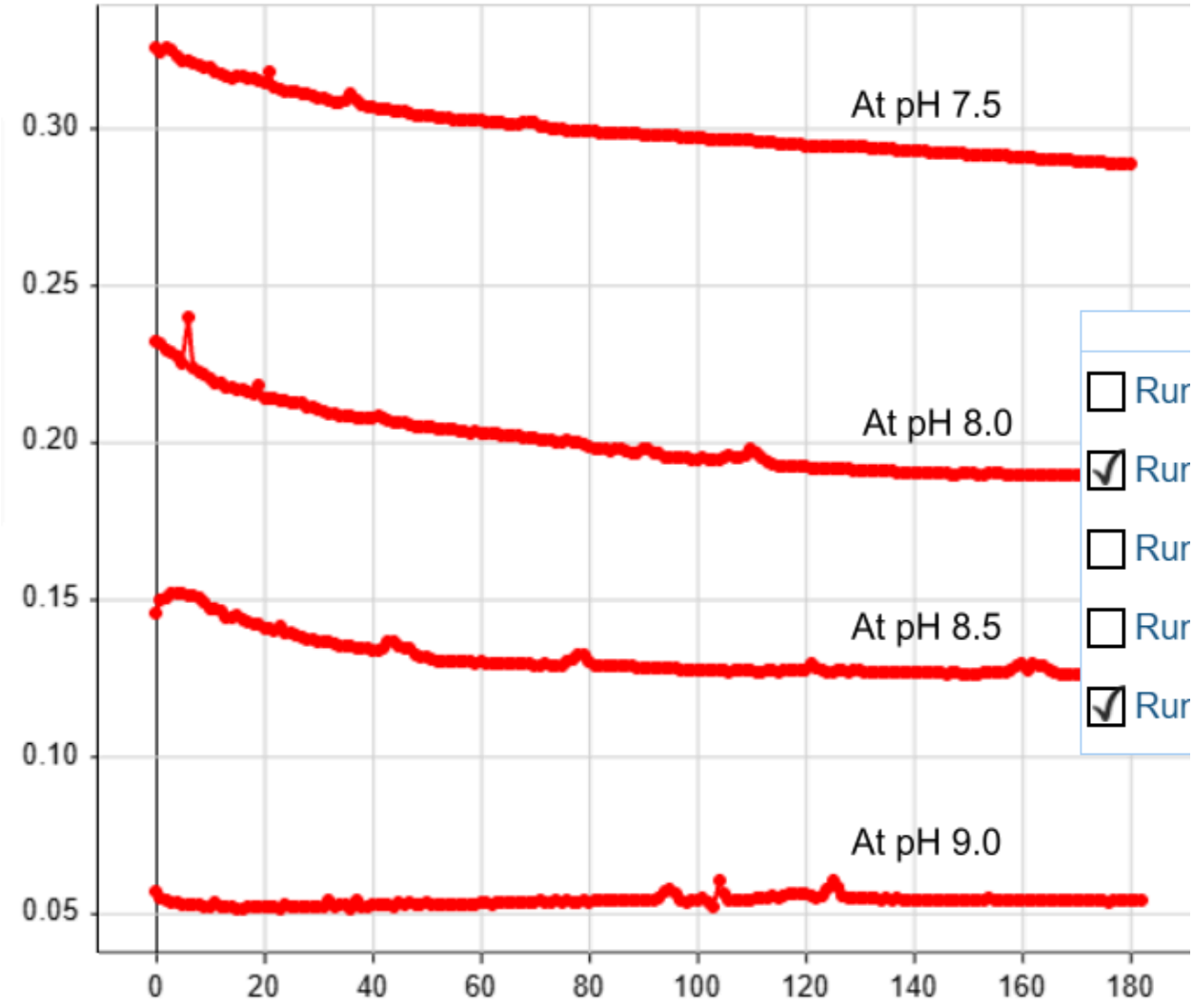
Graph 2: at pH 8.0



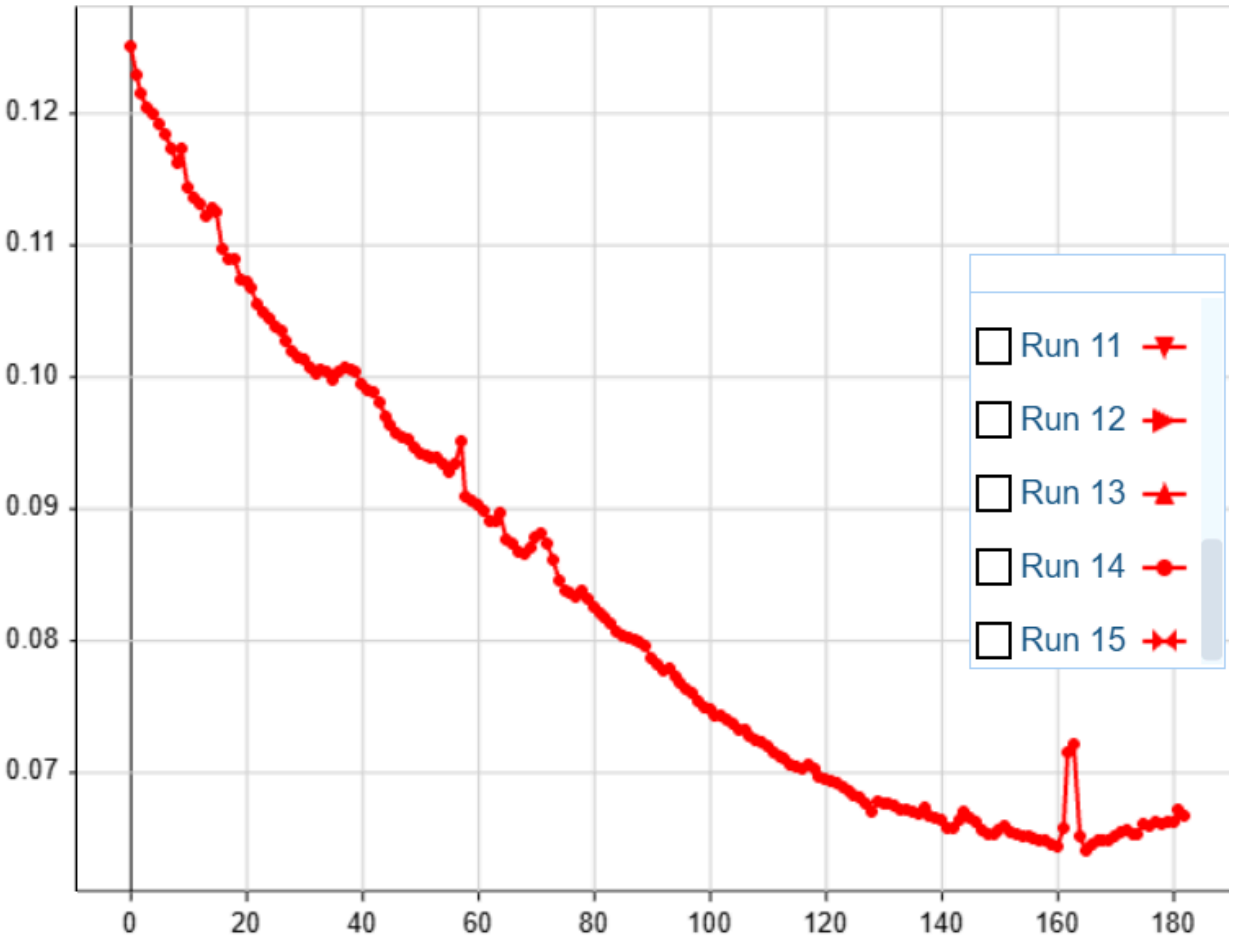
Graph 3: at pH 8.5



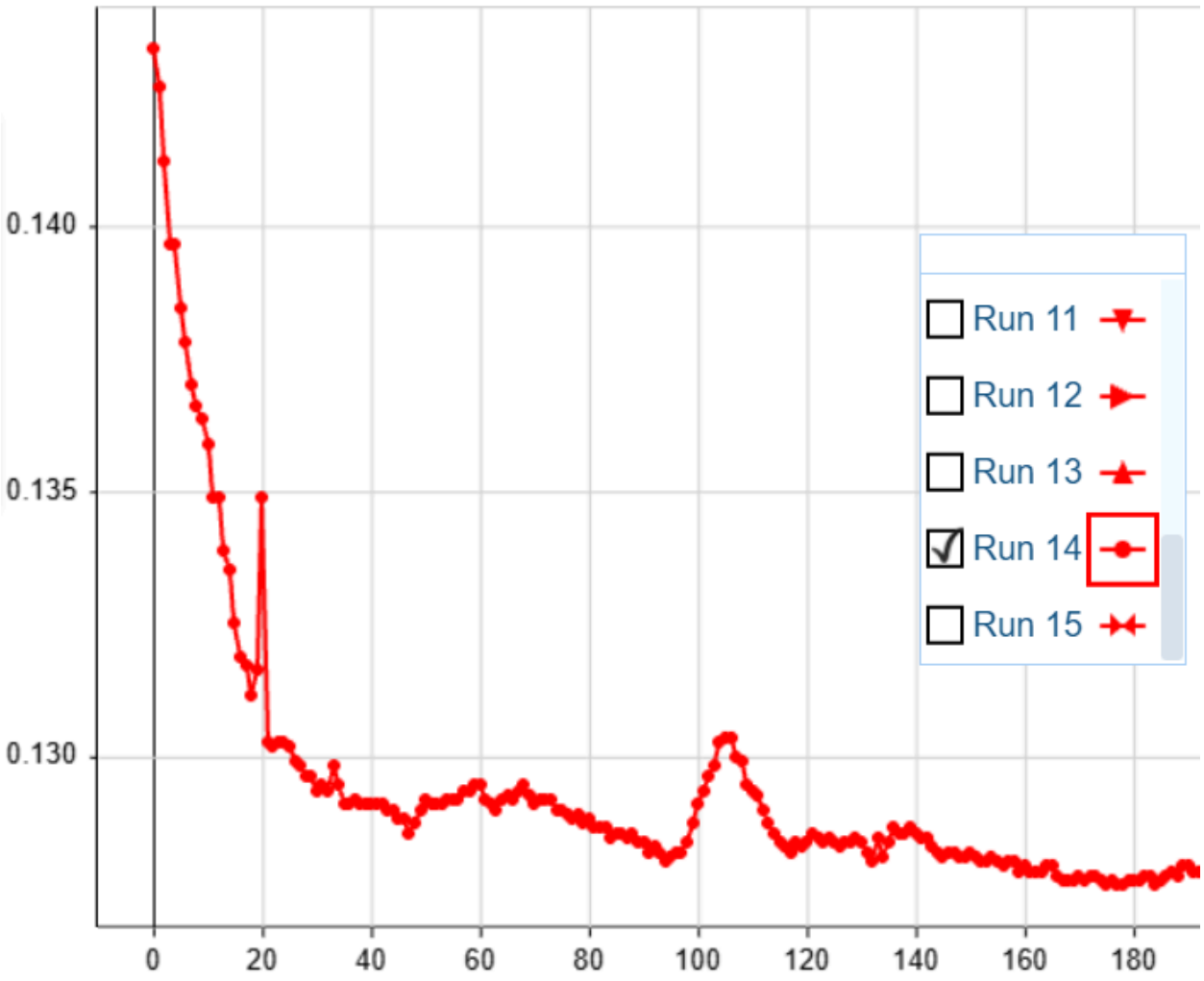
Graph 4: at pH 9.0



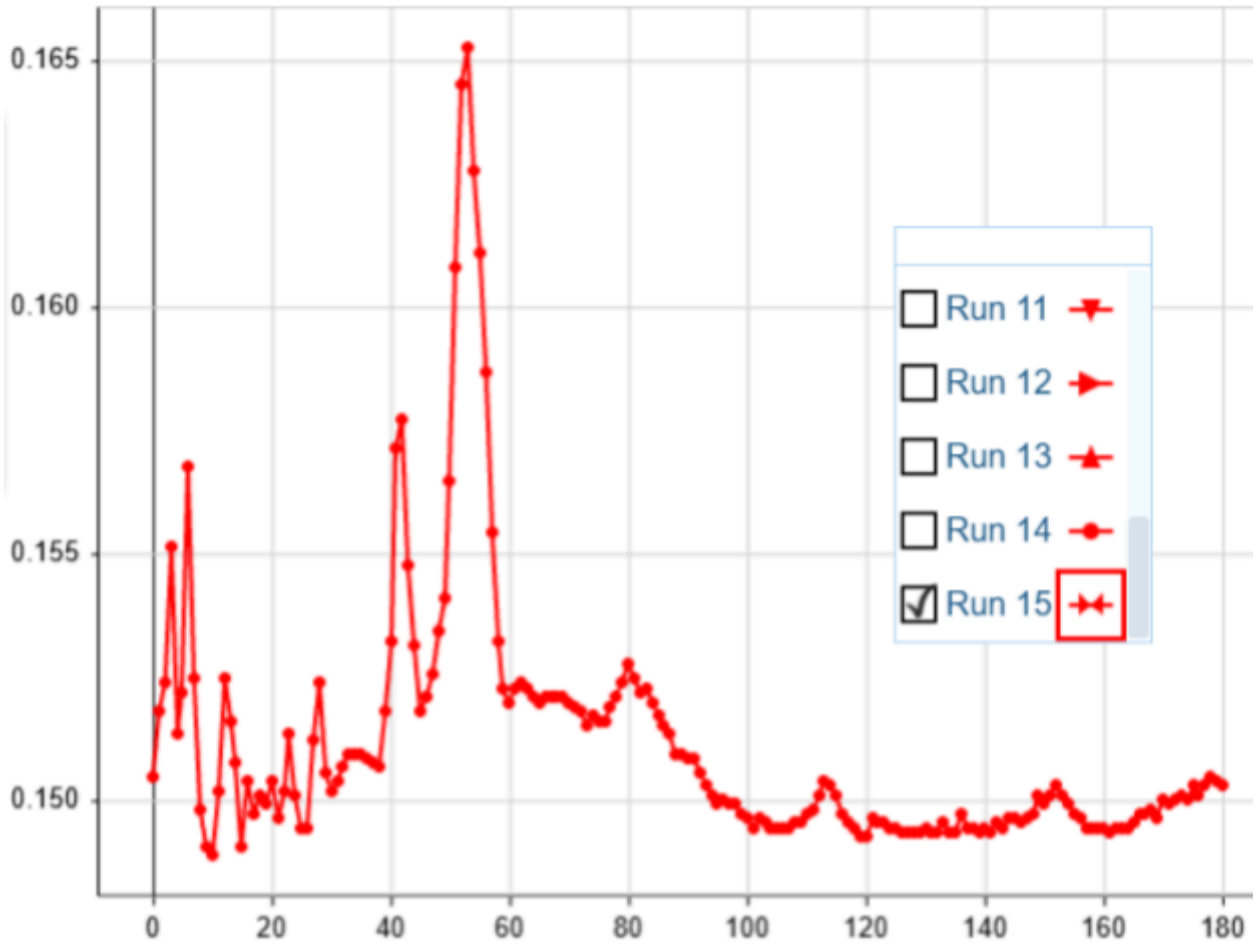
Graph 5: from pH 7.5~9.0



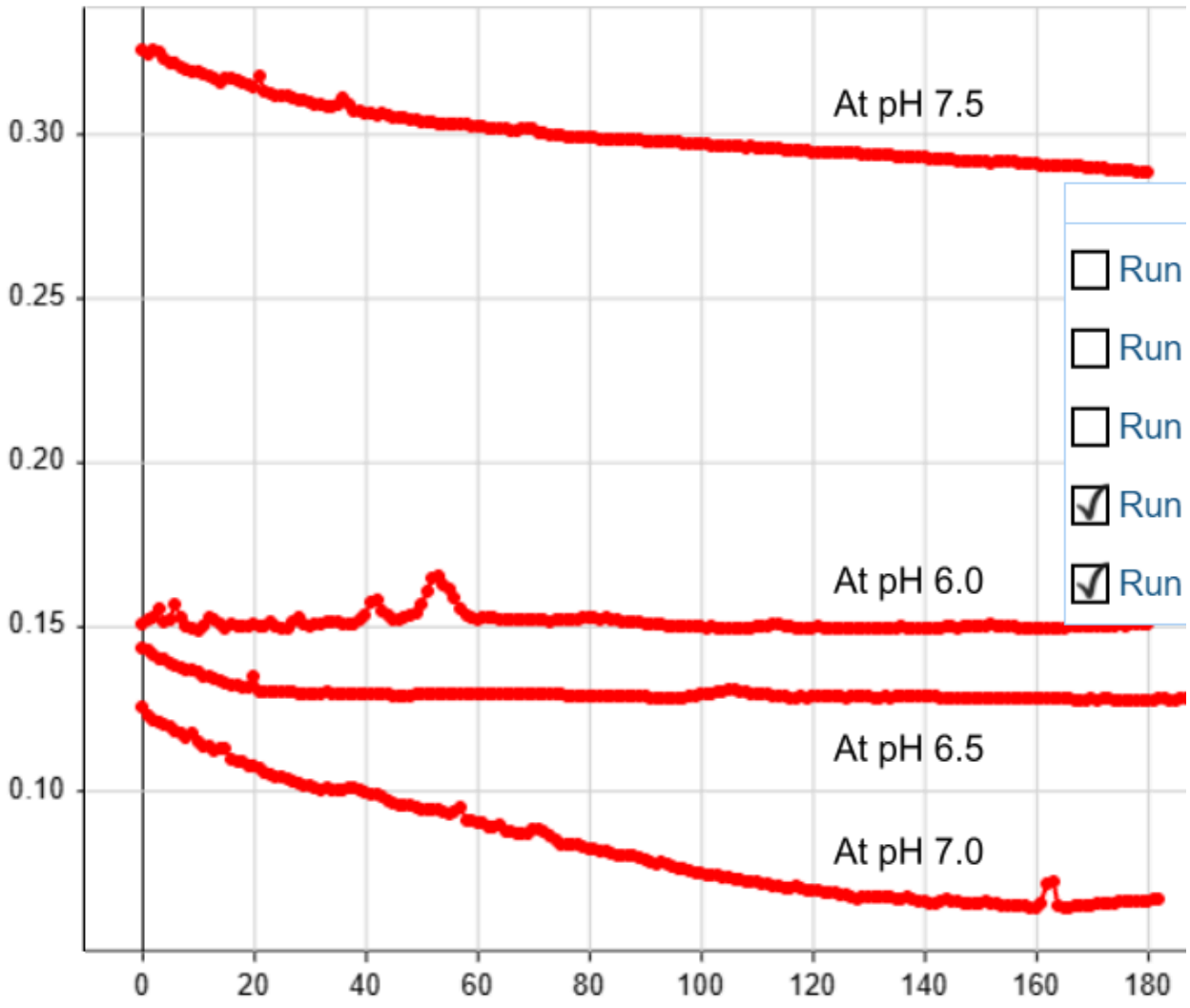
Graph 6: at pH 7.0



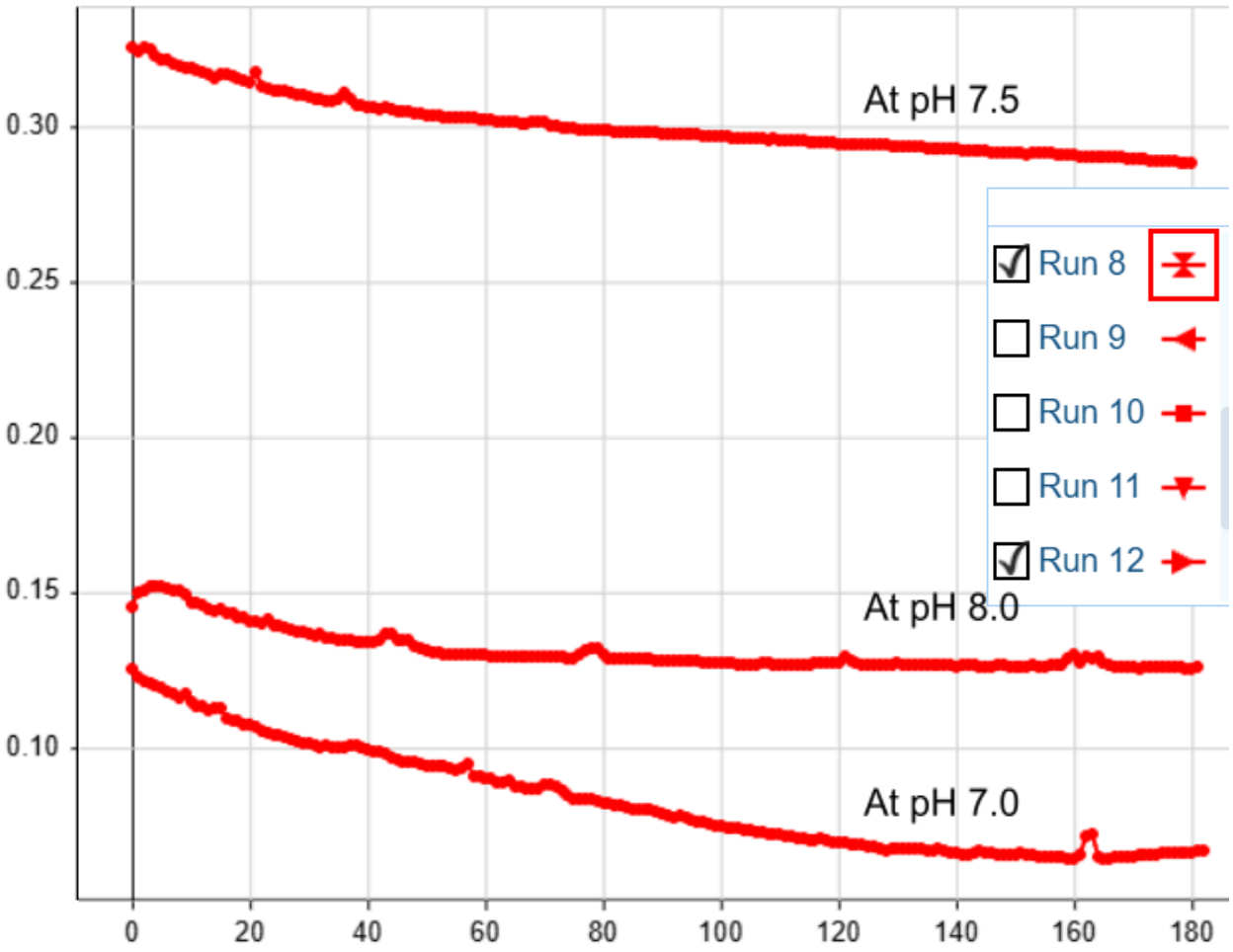
Graph 7: at pH 6.5



Graph 8: at pH 6.0



Graph 9: from pH 6.0~7.5



Graph 10: from pH 7.0~8.0

6.2 Analysis

All graphs showed different reaction rates of the oxidation of indigo carmine, presented as decreasing trends of measured red light absorbance at a peak wavelength of 650 nm of indigo carmine. When the pH level is increased and decreased from the median, the rate at which the absorbance vs time curve declines becomes shallower, meaning that the rate of the reaction decreases.

The average rate of decrease in absorbance over the same early time interval, from 11 s to 60 s, was the fitted slope between two end data points and calculated by linear regression.

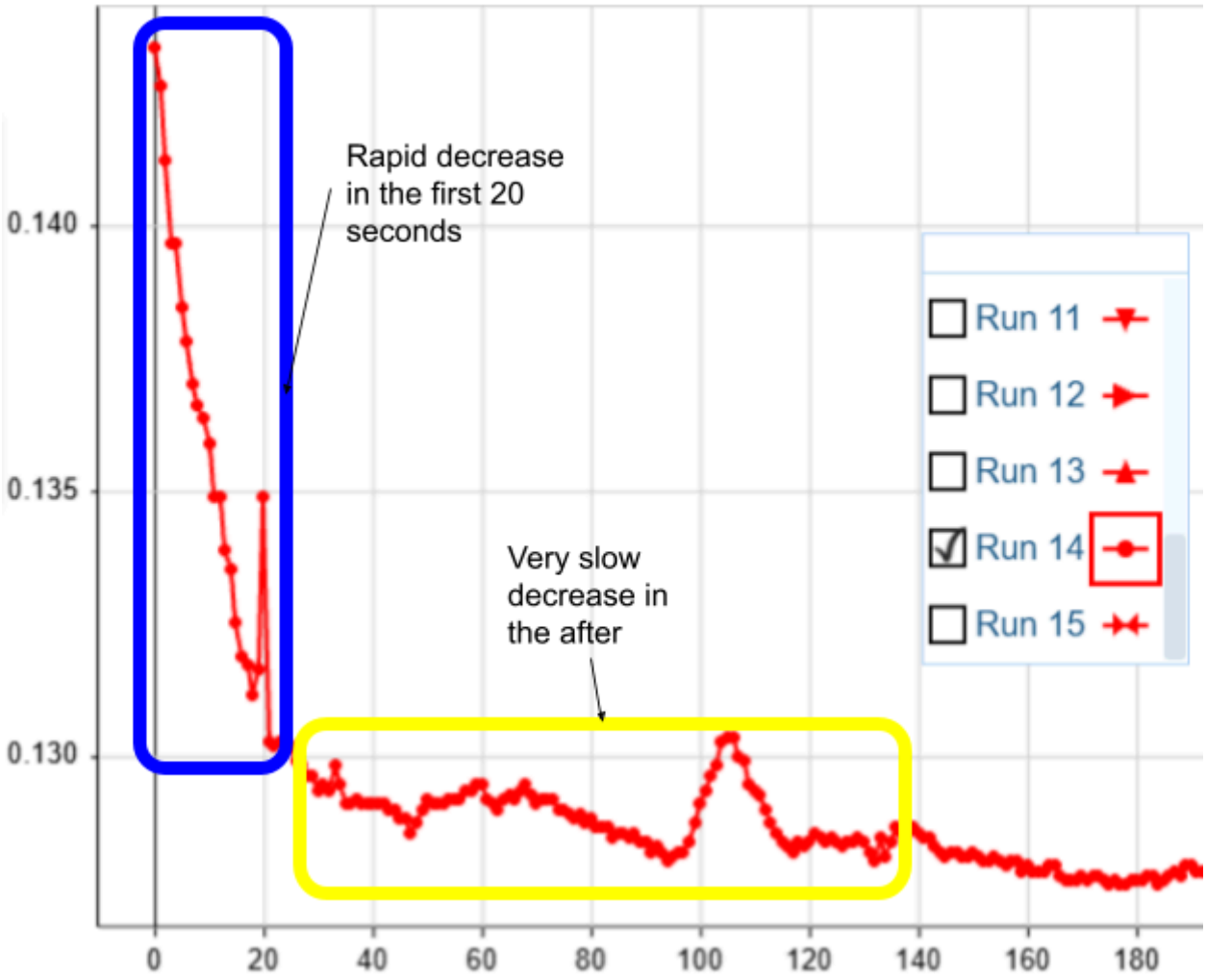
	pH 7.5 (11 s ~ 60 s)	pH 7.0 (11 s ~ 60 s)	pH 6.5 (11 s ~ 60 s)	pH 6.0 (11 s ~ 35 s)
Slope	-0.000337335	-0.000433277	-0.0000939736	0.000119481

	pH 7.5 (11 s ~ 60 s)	pH 8.0 (11 s ~ 60 s)	pH 8.5 (11 s ~ 60 s)	pH 9.0 (11 s ~ 60 s)
Slope	-0.000337335	-0.000320768	-0.000325474	0.0000257863

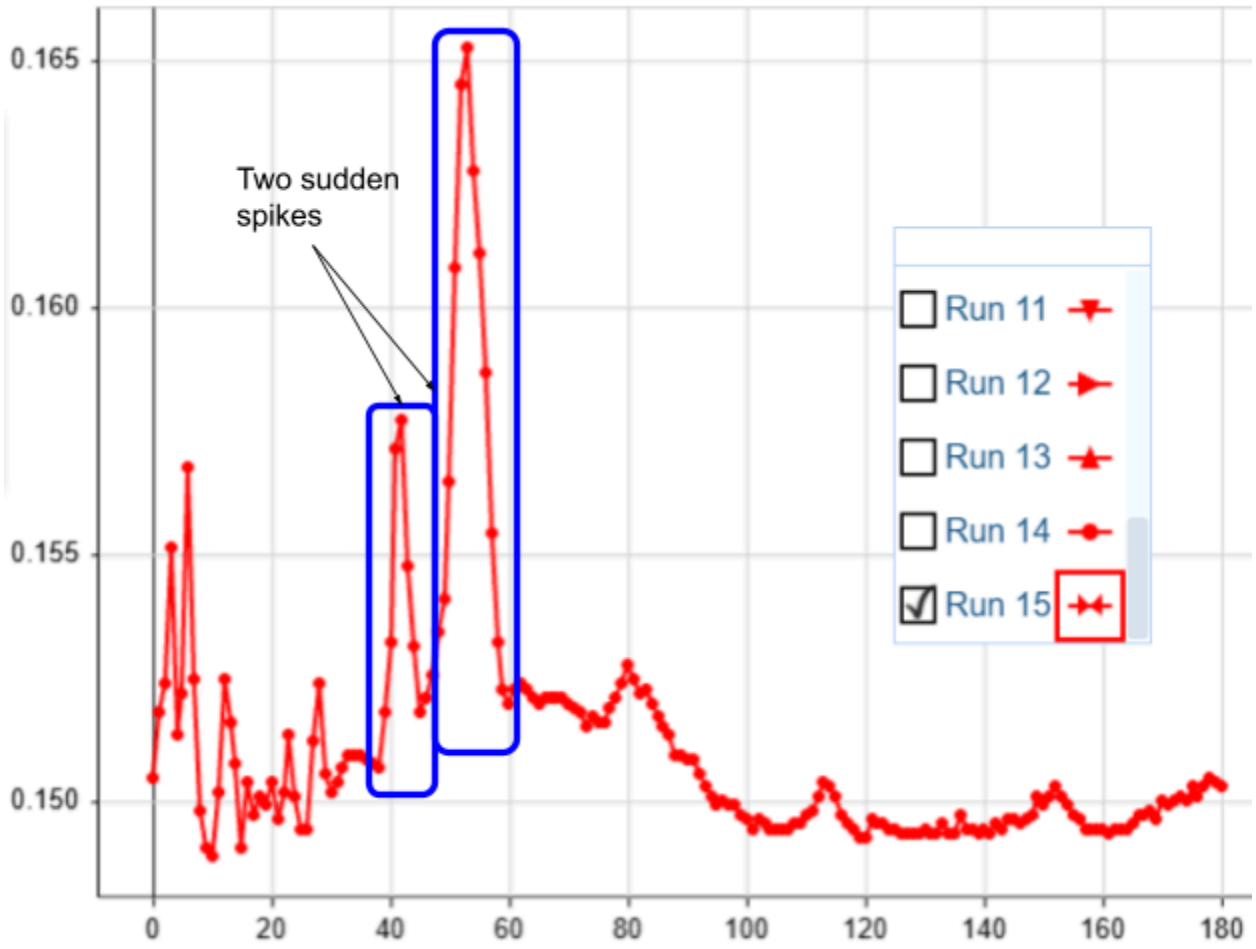
The data in the first 10 seconds interval is cut because it is when diffusion and oxidation happen at the same time. The data may not represent the real reaction rate. For pH 6.0, a significant disturbance occurred between 40 s and 60 s, so the regression is restricted to 11 s to 35 s to avoid factors that are not related to oxidation kinetics.

All the calculated rates show a clear dependence on pH level of the sample. The steepest negative slope occurs at pH 7.0, about $-\ 0.000433$ absorbance per second, followed by slightly shallower slopes between pH 7.5 to 8.5. From pH 8.0 to pH 8.5, the slope becomes steeper but the difference is very small, so it nearly demonstrates an increasing trend of slope magnitude. At pH 9.0, the slope is positive and has the smallest absolute value, about 2.6×10^{-5} absorbance per second, compared to other slopes. It indicates minimal oxidation measured within all samples. On the other hand, the slope is also positive at pH 6.0 and all slopes from pH 7.0 are consistently becoming shallower. At pH 6.5, the graph shows a rapid decrease in absorbance over the first 20 seconds of measurement, but immediately after the

decrease becomes very slow(see graph 11). At pH 6.0, two obvious spikes occurred between 35 s and 60 s, therefore this interval is excluded from the linear regression analysis because of its huge disturbance in data compared to others(see graph 12). At two extreme pH values, the slopes are positive meaning increasing absorbance over time, this suggests that the oxidation effect is so small at these two pH levels that noise or uncertainties within the colorimeter dominate over.



Graph 11: At pH 6.5, a huge difference occurs between the rates of decrease in the blue region(first 20 seconds) and in the yellow region(after 20 seconds).



Graph 12: At pH 6.0, two sudden spikes occur between 35 seconds and 60 seconds.

7. Evaluation

This experiment produces consistent trends in the absorbance-time relationship. It indicates that it is suitable and reliable for comparing the oxidation kinetics of chlorine sanitizer at different pH levels. However, there is a contradiction between the hypothesis and the data at pH 6.0~6.5.

After my investigation, I found that this is likely caused by the formation of Cl_2. In each sample, there are always chloride ions present from multiple sources. For example, when adjusting pH

by adding diluted hydrochloric acid, chloride ions were introduced. And the manufacturing process of bleach naturally produces chloride ions as byproducts(CLOROX, n.d.):

$$Cl_2 + 2OH^- \rightarrow OCl^- + Cl^- + H_2O$$

All these sources lead to the presence of chloride ions. They have reaction with hypochlorous acid:

$$HOCl(aq) + H^+(aq) + Cl^-(aq) \Leftrightarrow Cl_2(g) + H_2O(l)$$

According to Le Chatelier's Principle, at lower pH levels, the concentration of H^+ is high, and with an already high amount of chloride ions, the reaction shifts to the product side, which favors the production of chlorine gas.(Mehendale. F V., 2020) Chlorine gas is volatile, so they can easily escape from the beaker and cause further shifting in the equilibrium. It ultimately causes inevitable loss in the oxidation agent. Therefore, even if the hypochlorous acid concentration always dominates over hypochlorite ion concentration at lower pH levels, the formation of chlorine gas leads to a net loss in total chlorine available to react with indigo carmine. On the other hand, when pH levels are high, this reaction is merely side reaction due to limited amount of H^+. Moreover, hypochlorite ions are relatively stable in basic solutions, which makes the effect of chlorine gas formation negligible at higher pH levels.

The limitation of the analysis method is that the rates measured at two extreme pH values carry lower statistical confidence due to their significantly lower coefficient of determination value, $R^2 \leq 0.5$. At the middle three pH levels, the R^2 values are around 0.95. This difference means that the apparent linear model does not explain the absorbance-time relationship under the

extreme conditions in this experiment. The same point is demonstrated by the positive slopes in both graphs. Thus, the calculated rates at these two values have lower reliability and should be treated as approximate indications of the general trend rather than precise numerical values.

The major limitation of this experiment is the short delay between adding diluted indigo carmine solution and measuring absorbance with the colorimeter for each trial. It not only causes difference in starting absorbance, but more importantly, the unobserved rate of reaction at the beginning of oxidation. This limitation doesn't invalidate the trends observed, but causes, at the time periods when the rates are steepest, the data were lost prior to colorimeter measurement. Future investigations could minimize this effect by adding the dye solution directly to the cuvette or using more professional equipment that can mix the solutions sufficiently while collecting the data during this process. Furthermore, additional redox titration for measuring chlorine content can also be applied to confirm the conclusion of chlorine loss at lower pH levels.

Overall, the samples with the pH values between 7.0~8.0 show the highest rate of oxidation. This outcome aligns the practical swimming pool standard, which is maintaining the pH level between 7.4~7.6. It differs slightly from the theoretical chlorine effectiveness because both the $HOCl/OCl^-$ ratio and total chlorine content were not directly measured in this experiment. Consequently, the result is more practical rather than theoretical in measuring chlorine sanitizer effectiveness.

8. Conclusion

Through this investigation, I found that the oxidation kinetics of indigo carmine by chlorine sanitizer in simulated pool water depends on the pH level. The highest oxidation rate occurs in a neutral pH range of 7.0~8.0. As pH increases or decreases from this mean range, the rates of decline in the absorbance vs time curve become slower. It is because a higher proportion of less effective oxidizing agents at higher pH and reduced total chlorine availability at lower pH. This answers the main research question on this topic. Additionally, it also explains a key reason why all swimming pools maintain a nearly neutral pH level: to maximize the practical chlorine effectiveness.

9. Bibliography

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